



CANDIDATE
NAME

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CENTRE
NUMBER

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CANDIDATE
NUMBER

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9702/31

October/November 2022

2 hours

You must answer on the question paper.

You will need: The materials and apparatus listed in the confidential instructions

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You will be allowed to work with the apparatus for a maximum of 1 hour for each question.
- You should record all your observations in the spaces provided in the question paper as soon as these observations are made.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 40.
- The number of marks for each question or part question is shown in brackets [].

For Examiner's Use	
1	
2	
Total	

This document has **12** pages.

You may not need to use all of the materials provided.

1 In this experiment, you will investigate oscillations of card shapes.

(a) You have been provided with a circular card of radius 10.0 cm.

- Draw a circle on the card of radius 9.0 cm, as shown in Fig. 1.1.

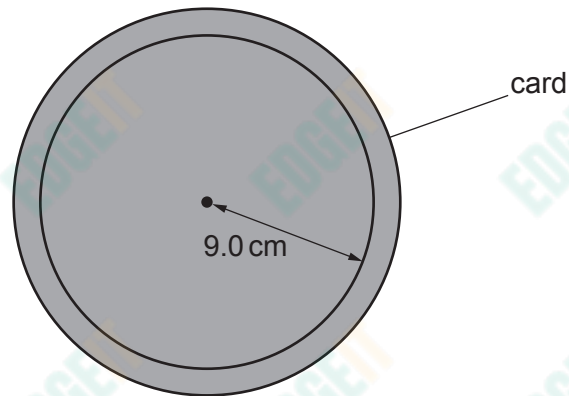


Fig. 1.1 (not to scale)

- Fold the card in half. Cut carefully along the line, as shown in Fig. 1.2, and keep both parts of the card.

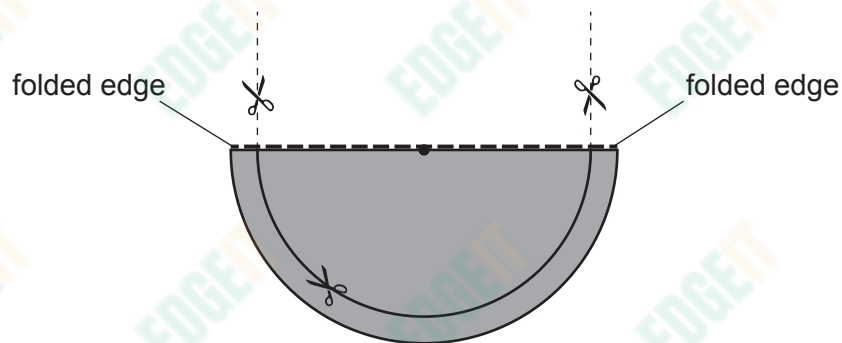


Fig. 1.2 (not to scale)

- The distance between the centre of one side of the card shape and the centre of the other side is d , as shown in Fig. 1.3.

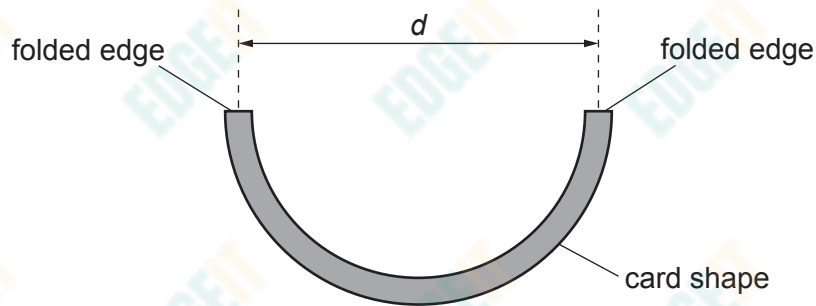


Fig. 1.3

Measure and record d .

$d = \dots\dots\dots$ [1]

- (b)
- Stand the card shape on the bench.
 - Adjust the loops of card until the distance between the points where the loops touch the bench is approximately 3 cm, as shown in Fig. 1.4.

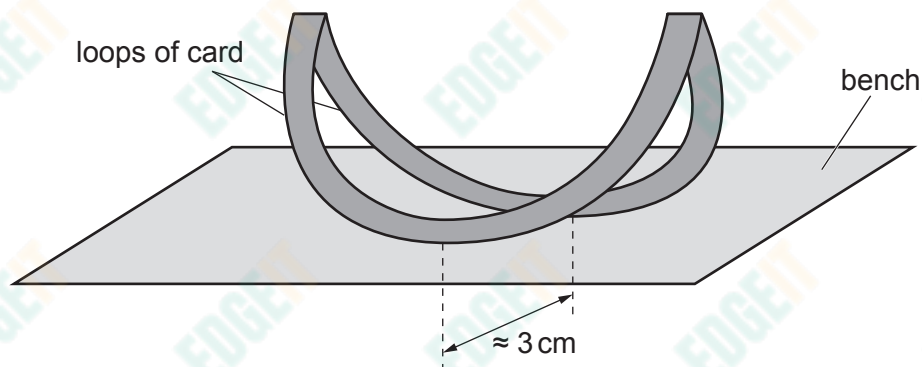


Fig. 1.4

- Gently press down one side of the card shape through a short distance. Release the card shape so that it oscillates.
- Determine the period T of these oscillations.

$T = \dots\dots\dots$ [2]

- (c) Use the remaining card to cut out shapes of smaller radius, each with the same width of 1.0 cm.

For each card shape, measure d and repeat (b). Repeat until you have five sets of values of d and T .

Record your results in a table. Include values of T^2 in your table.

[8]

- (d) (i) Plot a graph of T^2 on the y -axis against d on the x -axis.

[3]

- (ii) Draw the straight line of best fit.

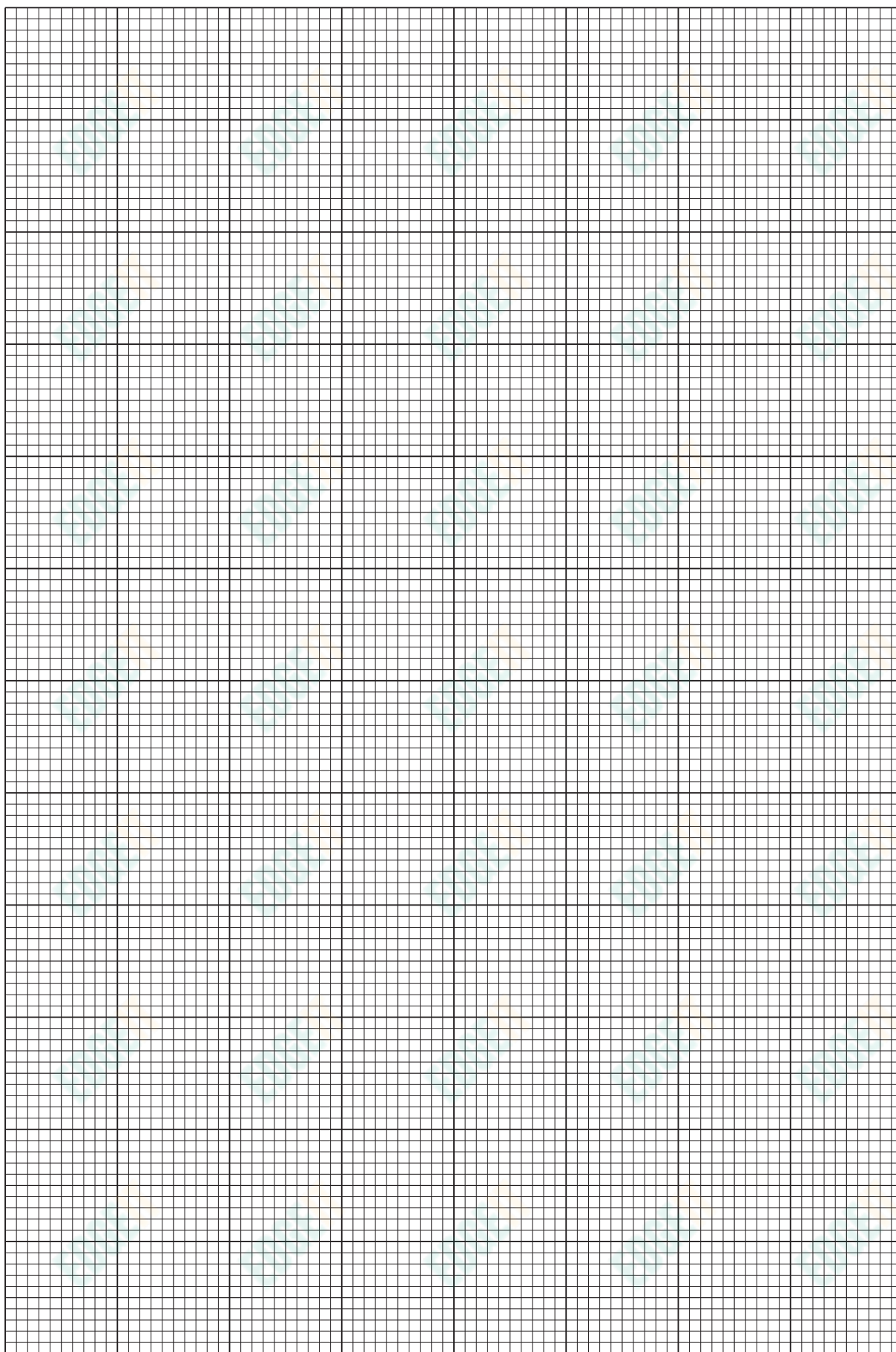
[1]

- (iii) Determine the gradient and y -intercept of this line.

gradient =

y -intercept =

[2]



- (e) It is suggested that the quantities T and d are related by the equation

$$T^2 = Ad + B$$

where A and B are constants.

Using your answers in (d)(iii), determine the values of A and B .
Give appropriate units.

$$A = \dots\dots\dots$$

$$B = \dots\dots\dots$$

[2]

- (f) Theory suggests that

$$A = \frac{2\pi^2}{g}$$

where g is the acceleration of free fall.

Use your value of A in (e) to determine a value for g .
Give an appropriate unit.

$$g = \dots\dots\dots$$

[1]

[Total: 20]

You may not need to use all of the materials provided.

2 In this experiment, you will investigate the collision of two pendulums.

- (a) (i) • Mould the two pieces of modelling clay onto the ends of the **shorter** strings to make two pendulums, as shown in Fig. 2.1.

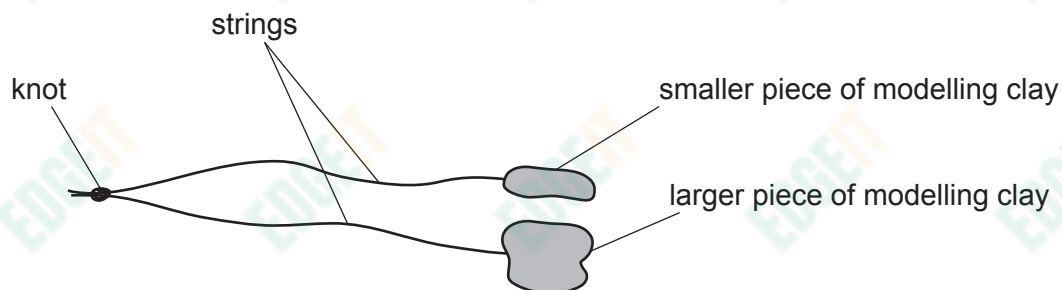


Fig. 2.1

- Set up the apparatus as shown in Fig. 2.2.

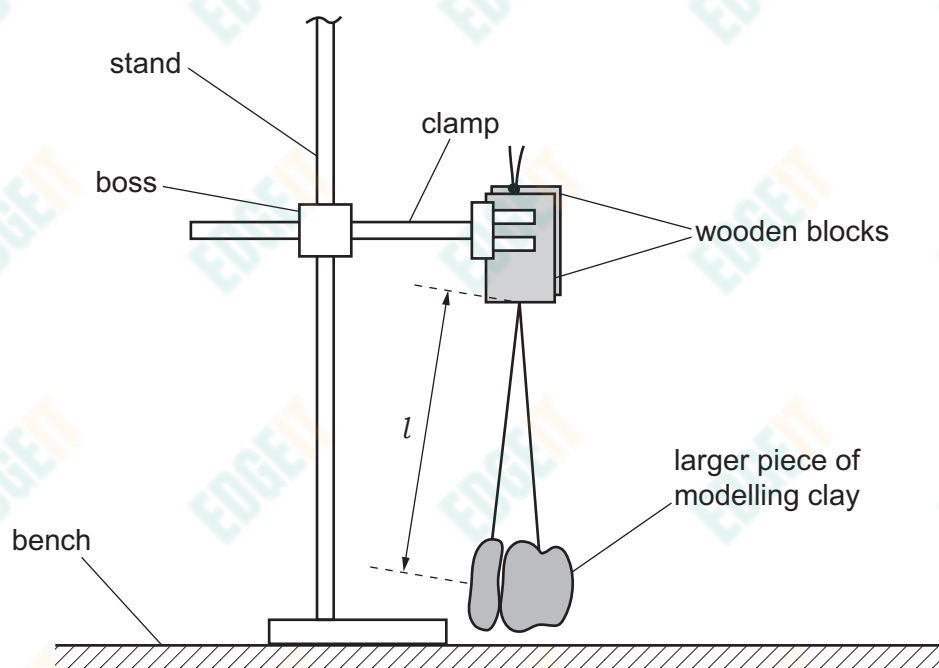


Fig. 2.2

- Adjust the modelling clay until the lengths of the pendulums are the same.
- The distance between the bottom of the wooden blocks and the centre of the smaller piece of modelling clay is l , as shown in Fig. 2.2.

Measure and record l .

$l =$ [1]

- (ii) • Hold the larger pendulum a short distance away from the smaller pendulum, as shown in Fig. 2.3.

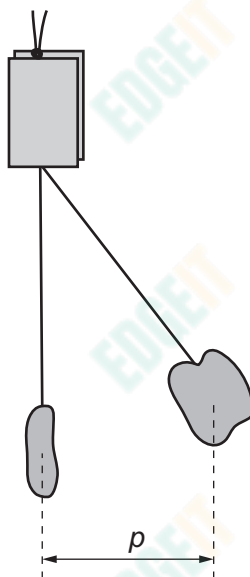


Fig. 2.3

- The horizontal distance between the centres of the pendulums is p .
- Hold the larger pendulum so that p is approximately 12 cm.
- Measure and record p .

$p =$ [1]

- (iii) Calculate R where

$$R = \sqrt{\left(1 - \frac{p^2}{l^2}\right)}.$$

$R =$ [1]

- (b) Justify the number of significant figures that you have given for your value of R .

.....

 [1]

- (c) (i)
- Hold the larger pendulum so the horizontal distance between the centres of the pendulums is p .
 - Release the larger pendulum so that the pendulums collide.
 - After colliding, the maximum angle between the vertical and the string of the **smaller** pendulum is θ , as shown in Fig. 2.4.

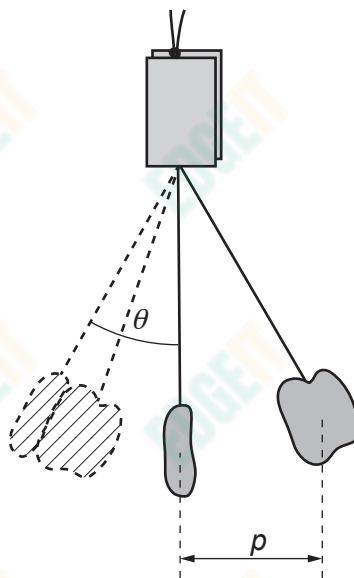


Fig. 2.4

Measure and record θ .

$\theta = \dots\dots\dots^\circ$ [2]

- (ii) Estimate the percentage uncertainty in your value of θ . Show your working.

percentage uncertainty = $\dots\dots\dots\%$ [1]

- (d) • Remove the modelling clay from the strings.
- Repeat (a) and (c)(i) using the **longer** strings.

$l =$

$p =$

$R =$

$\theta =$ [°]

[3]

- (e) It is suggested that the relationship between θ and R is

$$k(1 - \cos \theta) = 1 - R$$

where k is a constant.

Using your data, calculate two values of k .

first value of k =

second value of k =

[1]

- (f) It is suggested that the percentage uncertainty in the values of k is 10%.

Using this uncertainty, explain whether your results support the relationship in (e).

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..... [1]

- (g) (i) Describe **four** sources of uncertainty or limitations of the procedure for this experiment.

For any uncertainties in measurement that you describe, you should state the quantity being measured and a reason for the uncertainty.

1

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2

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3

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4

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[4]

- (ii) Describe **four** improvements that could be made to this experiment. You may suggest the use of other apparatus or different procedures.

1

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2

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3

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4

.....

[4]

[Total: 20]

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